

17. GRE Scores The quantitative reasoning scores on the Graduate Record Examination (GRE)[®] were recorded for students admitted to three different graduate programs at a local university.

Graduate Program					
1		2		3	
Life Sciences		Physical Sciences		Social Sciences	
630	660	660	760	440	540
640	660	640	670	330	450
470	480	720	700	670	570
600	650	690	710	570	530
580	710	530	450	590	630

- Do these data provide sufficient evidence to indicate a difference in the mean GRE scores for applicants admitted to the three programs?
- Find a 95% confidence interval for the difference in mean GRE scores for Life Sciences and Physical Sciences.
- If you find a significant difference in the average GRE scores for the three programs, use Tukey's

b. 95% CI.

$$\begin{aligned} \bar{X}_1 - \bar{X}_2 &\pm t_{0.025, 27} \sqrt{MSE \left(\frac{1}{10} + \frac{1}{10} \right)} \\ &= 608 - 653 \pm 2.052 \\ &= (-128.946, 38.946) \end{aligned}$$

CI include 0, we don't have sufficient evidence to say there's significant difference between μ_1 and μ_2 .

$$\begin{aligned} c. \quad w &= q_{0.05}(3, 27) \sqrt{\frac{MSE}{10}} \\ &= 3.506 \sqrt{\frac{8367.78}{10}} = 101.4185 \end{aligned}$$

$|\bar{X}_1 - \bar{X}_2| = 45 < w \Rightarrow \mu_1$ and μ_2 aren't significant different

$|\bar{X}_1 - \bar{X}_3| = 176 < w \Rightarrow \mu_1$ and μ_3 aren't significant different

$|\bar{X}_2 - \bar{X}_3| = 110 > w \Rightarrow \mu_2$ and μ_3 are significant different.

$$\begin{aligned} a. \quad \sum X_1 &= 6080 & \sum X_2 &= 6530 & \sum X_3 &= 5320 \\ \sum X_1^2 &= 3752000 & \sum X_2^2 &= 4343700 & \sum X_3^2 &= 2921200 \end{aligned}$$

$$CM = \frac{G^2}{n} = \frac{(6080 + 6530 + 5320)^2}{30} = 10716163.33$$

$$SS = \sum X^2 - CM = 3752000 + 4343700 + 2921200 - 10716163.33 = 300736.67$$

$$SST = \sum \frac{T_i^2}{n_i} - CM = \frac{6080^2}{10} + \frac{6530^2}{10} + \frac{5320^2}{10} - CM = 74806.67$$

$$SSE = SS - SST = 225930$$

ANOVA	SS	df	MS	F
SST	74806.67	2	37403.335	4.47
SSE	225930	27	8367.78	
SS	300736.67	29		

$$H_0: \mu_1 = \mu_2 = \mu_3 \quad H_a: \text{at least one mean is different}$$

$$\alpha = 0.05$$

$$F_{STAT} = \frac{MST}{MSE} \sim F_{(2, 27)}$$

$$F^* = 4.47$$

$$RR: F_{0.05}(2, 27) = 3.35$$

$$\{ F \mid |F^*| > 3.35 \}$$

$\therefore 4.47 > 3.35 \Rightarrow \text{Reject } H_0 \Rightarrow \text{we don't have sufficient evidence to say that means are equal.}$